# Cross-Domain Complaint Understanding via Multilingual Encoders

## B.Tech AI/ML Capstone Project Proposal

**Institution:** IIIT Nagpur
**Project Type:** NLP Research + Working System
**Primary Research Area:** Multilingual NLP, Transfer Learning, Complaint Understanding
**Proposed Duration:** 16 weeks
**Primary Model:** XLM-R
**Supporting Models:** mT5 for summarization
**Explainability:** SHAP
**Retrieval:** FAISS
**Deployment:** FastAPI + lightweight web interface

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## 1. Executive Summary

Organizations receive complaints through customer-support systems, government portals, educational institutions, banking services, telecom platforms, and e-commerce systems. These complaints are frequently written in different languages, informal language, transliterated text, and code-mixed forms such as Hinglish.

A complaint-understanding system therefore needs to do more than classify English text within a single domain. It should ideally recognize the underlying intent of a complaint even when the wording, language, and application domain change.

This project proposes a focused multilingual NLP system for **cross-domain complaint understanding**. The central research question is whether a multilingual transformer encoder can learn complaint representations that transfer across domains and languages with limited labeled data.

The system will use a multilingual language-identification component, an XLM-R encoder for complaint representation and intent classification, semantic retrieval for finding related complaints, mT5-based summarization, and SHAP-based explanations of intent predictions.

The project deliberately avoids adding an LLM reasoning layer, RAG, NER, urgency prediction, or systemic-failure detection. These are outside the core research question and can be considered future extensions.

The final deliverable will be both:

1. **A research study** evaluating cross-domain and cross-language transfer; and
2. **A working complaint-analysis application** that accepts a complaint and returns its language, domain/intent, similar complaints, concise summary, and explanation of the classification.

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# 2. Problem Statement

Existing complaint-processing systems are commonly designed around a particular organization or domain. A model trained on banking complaints may perform well on banking data but may not transfer effectively to telecom, education, government, or e-commerce complaints.

A second challenge is multilingual and code-mixed communication. Indian users may submit complaints in English, Hindi, or Hinglish, for example:

> "Mera recharge successful hai but internet abhi tak activate nahi hua."

A system should ideally understand that this has the same underlying intent as:

> "My recharge was successful but the internet service has not been activated."

The project therefore investigates whether a **shared multilingual representation** can capture complaint intent independently of the specific domain and language.

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# 3. Research Question

## Primary Research Question

> **Can multilingual transformer encoders learn domain-independent representations of complaints that generalize across unseen domains and languages under limited labeled-data conditions?**

## Secondary Research Questions

### RQ1 — Cross-domain transfer

How much does complaint-intent performance degrade when a model trained on some complaint domains is evaluated on a previously unseen domain?

### RQ2 — Cross-language transfer

How well does the learned representation transfer between English, Hindi, and Hinglish?

### RQ3 — Contrastive representation learning

Can contrastive training improve the alignment of semantically equivalent complaints expressed using different wording, languages, or domains?

### RQ4 — Semantic retrieval

Can the learned complaint representation retrieve related complaints more effectively than traditional lexical similarity methods?

### RQ5 — Summarization

Can a multilingual pretrained summarization model produce concise and factually consistent summaries of complaint narratives?

### RQ6 — Explainability

Can SHAP identify meaningful textual features associated with the model's intent predictions?

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# 4. Hypotheses

The project will test the following hypotheses rather than assuming that the desired results will occur.

### H1 — Multilingual transfer

A multilingual encoder such as XLM-R will provide stronger cross-language complaint-intent transfer than a comparable English-only encoder.

### H2 — Cross-domain transfer

A model trained jointly on multiple complaint domains will generalize better to an unseen complaint domain than a model trained exclusively on one domain.

### H3 — Contrastive learning

Contrastive fine-tuning will improve semantic alignment between complaints expressing similar intents across different languages and domains.

### H4 — Semantic retrieval

XLM-R representations trained with a similarity objective will outperform a TF-IDF cosine-similarity baseline for retrieving semantically related complaints.

### H5 — Code-mixed robustness

Including naturally occurring Hindi-English code-mixed examples during training or adaptation will improve performance on Hinglish complaints compared with an English-only training regime.

### H6 — Summarization

A pretrained multilingual mT5-based summarization model will produce useful complaint summaries, while complaint-specific fine-tuning on a smaller annotated summary set may improve factuality and relevance.

### H7 — Explanation faithfulness

The tokens identified as influential by SHAP will correspond to meaningful textual evidence for a substantial proportion of intent predictions.

**No numerical outcome is assumed by these hypotheses.**

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# 5. Proposed System

The final system will follow this pipeline:

```
                  ┌──────────────────┐
                  │    COMPLAINT     │
                  └────────┬─────────┘
                           │
                           ▼
               ┌───────────────────────┐
               │ Language Identifier   │
               │ EN / HI / Hinglish    │
               │ / Other               │
               └──────────┬────────────┘
                          │
                          ▼
             ┌─────────────────────────┐
             │      XLM-R Encoder      │
             │                         │
             │ Multilingual +          │
             │ Cross-domain learning   │
             └────────────┬────────────┘
                          │
           ┌──────────────┼──────────────┐
           │              │              │
           ▼              ▼              ▼
   ┌────────────┐ ┌─────────────┐ ┌─────────────┐
   │   Intent   │ │  Semantic   │ │ Summarizer  │
   │ Classifier │ │  Retrieval  │ │     mT5     │
   └─────┬──────┘ └──────┬──────┘ └──────┬──────┘
         │               │               │
         ▼               ▼               ▼
       Intent      Similar Cases      Summary
         │
         ▼
  ┌────────────────┐
  │      SHAP      │
  │   Explanation  │
  └───────┬────────┘
          │
          ▼
┌─────────────────────────┐
│      Final Output       │
│ Language                │
│ Domain / Intent         │
```

```
          │ Confidence              │
          │ Similar complaints      │
          │ Complaint summary       │
          │ Explanation             │
          └─────────────────────────┘
```

---

# 6. Component 1 — Language Identification

The first stage identifies the language characteristics of the incoming complaint.

The initial application will focus on:

- English
- Hindi
- Hinglish/code-mixed Hindi-English
- Other/unsupported

The language-identification component will not attempt to solve general-purpose language identification for every Indian language. Its purpose is to provide reliable routing and analysis for the languages represented in the project's complaint dataset.

### Proposed implementation

Possible approaches will be evaluated during the baseline stage:

- fastText-based language identification
- lightweight multilingual transformer classifier
- XLM-R/MuRIL-based classifier where appropriate

Public Indian-language identification resources are available for pretraining/evaluation. These include datasets containing large numbers of Indian-language examples, although they are not themselves complaint datasets.

### Deliverables

- Language-ID preprocessing module
- Language distribution analysis
- Accuracy/F1 evaluation
- Confusion matrix
- Evaluation on naturally occurring complaint text
- Hinglish/code-mixed error analysis

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# 7. Component 2 — Multilingual Complaint Encoder

## Core Model: XLM-R

XLM-R will be the central representation-learning model.

The model will be evaluated in several configurations:

1. XLM-R pretrained/frozen representation + classifier
2. Fine-tuned XLM-R
3. XLM-R with class-balancing strategies where necessary
4. XLM-R with contrastive learning

The goal is not to train a multilingual language model from scratch.

The project will fine-tune an existing multilingual encoder on complaint data.

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# 8. Complaint Intent Classification

The primary supervised task will be **complaint-intent classification**.

Instead of preserving unrelated labels from individual datasets, the project will define a common intent ontology.

A preliminary ontology may contain categories such as:

```
PAYMENT
 ├── payment_failed
 ├── payment_pending
 ├── payment_reversed
 └── charged_but_service_failed

SERVICE
 ├── service_unavailable
 ├── activation_failure
 └── service_quality

DELIVERY
 ├── delivery_delayed
 ├── delivery_failed
 └── incorrect_delivery

ACCOUNT
 ├── login_failure
```

```
 ├── account_access
 └── verification_failure

REFUND
 ├── refund_pending
 └── refund_failed

INFORMATION
 ├── status_request
 └── information_request
```

The final ontology will be determined only after examining the available datasets and their label distributions.

This is important because the project should not force incompatible labels from unrelated datasets into an artificial common taxonomy.

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# 9. Cross-Domain Experimental Design

Cross-domain transfer is the central research experiment.

Potential domains include:

- Banking
- Telecom
- E-commerce
- Education
- Government/public services

The exact final set will depend on the availability, quality, licensing, and compatibility of datasets.

## Experiment A — In-domain classification

Train and test within the same domain.

Purpose:

- establish normal supervised performance
- verify that the model learns the task
- provide a baseline for transfer experiments

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## Experiment B — Leave-one-domain-out transfer

For example:

```
TRAIN
Banking + Telecom + E-commerce + Education

             ↓

TEST
Government
```

The process will be repeated with different held-out domains where dataset sizes permit.

The important metric will be **macro-F1**, supplemented by per-class precision, recall, and F1.

---

### Experiment C — Limited-data adaptation

The unseen domain will initially receive zero labeled examples.

Then progressively small amounts of labeled data will be introduced:

```
0 examples
25 examples
50 examples
100 examples
250 examples
```

The purpose is to determine how quickly the multilingual representation adapts to a new domain.

These are experimental conditions, not expected performance claims.

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# 10. Cross-Language Experimental Design

The project will separately evaluate:

```
English → Hindi
English → Hinglish
Hindi → Hinglish
English + Hindi → Hinglish
```

where sufficient naturally occurring data exists.

The objective is to distinguish:

- multilingual transfer
- domain transfer

- code-mixing robustness

rather than treating all non-English examples as one category.

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# 11. Contrastive Learning

Contrastive learning will be the principal research extension beyond straightforward XLM-R fine-tuning.

The model will learn to bring semantically equivalent complaints closer in embedding space.

Example:

**Positive pair**

> "Mera paisa kat gaya but recharge nahi hua."

and

> "My account was charged but the recharge failed."

The model should learn to place these representations closer.

A hard negative may be:

> "Recharge failed and the money was automatically refunded."

This has related vocabulary but potentially a different intent.

The training strategy will therefore investigate:

- same-intent paraphrases
- cross-language pairs
- cross-domain pairs
- hard negatives

The final paper will compare standard XLM-R fine-tuning against XLM-R with the contrastive objective.

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# 12. Semantic Retrieval

The same complaint representation will be used to retrieve related complaints.

## Pipeline

```
Complaint
     ↓
```

```
XLM-R
    ↓
Complaint embedding
    ↓
FAISS index
    ↓
Top-k similar complaints
```

The baseline will be:

```
TF-IDF
+
cosine similarity
```

The proposed representation will be compared against this lexical baseline.

### Evaluation

Where manually annotated similarity pairs are available, evaluation will include:

- Precision@K
- Recall@K
- MRR
- qualitative error analysis

A small manually annotated evaluation set will be created if the public datasets do not provide suitable similarity labels.

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# 13. Complaint Summarization

Summarization is a supporting product capability rather than the central research contribution.

The purpose is to transform a long complaint into a short, actionable summary.

### Example

**Input**

> "I recharged my number yesterday for ₹399. The money was deducted from my account and the application says the recharge was successful, but I still don't have internet. I restarted my phone several times and contacted customer care twice but the problem has not been resolved."

**Expected output**

> "Customer reports that ₹399 was successfully charged for a recharge, but internet service remains unavailable despite contacting customer support twice."

The actual generated output will be evaluated rather than assumed to have this quality.

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## Model

A multilingual mT5-based summarization model will be evaluated first.

The project will not train a summarization model from scratch.

General multilingual summarization resources such as XL-Sum can be used for understanding/pretraining compatibility, while complaint-specific evaluation will be based on complaint text.

If resources permit, a smaller complaint-summary dataset will be manually created for domain adaptation.

### Proposed target

**300–500 manually written complaint-summary pairs** if feasible.

This is a target for annotation, not a claim that such a dataset already exists.

### Evaluation

Automatic:

- ROUGE-1
- ROUGE-2
- ROUGE-L
- BERTScore where appropriate

Human evaluation:

- factual consistency
- coverage of the main issue
- conciseness
- usefulness to a complaint handler

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# 14. SHAP Explainability

SHAP will be applied **only to the intent-classification component**.

It will not be used for language identification, semantic retrieval, or summarization.

For a prediction such as:

```
Intent:
payment_failure
```

```
Confidence:
0.91
```

the system will provide an explanation showing which parts of the complaint contributed to the prediction.

Example:

```
Complaint:
"Money was deducted but the recharge failed."

Important evidence:
"deducted"
"recharge"
"failed"
```

The exact explanation will be generated by the model and SHAP; these values are not predetermined.

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## SHAP evaluation

The project will investigate:

1. whether influential tokens correspond to meaningful evidence;
2. whether removing influential tokens changes model confidence;
3. whether explanations remain stable across languages;
4. whether explanations differ for correct and incorrect predictions.

We will **not** claim that SHAP improves human decision-making unless a properly designed human evaluation is actually conducted.

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# 15. Dataset Strategy

The project will not depend on the existence of a single large multilingual complaint dataset.

Instead, it will construct a unified research dataset from:

### A. Public complaint datasets

Potential sources will be selected based on:

- public accessibility
- licensing
- availability of complaint narratives
- domain relevance
- label quality

- language information
- compatibility with the common intent ontology

Banking complaint data is expected to provide one of the largest sources of real complaint narratives.

Other domains will be added where sufficient compatible data is available.

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### B. Indian multilingual data

Public multilingual Indian-language datasets will be used where appropriate for language and multilingual representation experiments.

However, general Indian-language datasets will **not** automatically be treated as complaint datasets.

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### C. Project-specific annotation

A smaller manually annotated Indian complaint set will be created to address the shortage of naturally occurring multilingual complaint data.

### Proposed target

Approximately:

**1,000–1,500 complaint examples**

subject to data access and collection feasibility.

The target will include a meaningful proportion of:

- English
- Hindi
- Hinglish

and multiple domains.

The exact distribution will be finalized after the public-data audit.

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# 16. Annotation Scheme

Each complaint in the project-specific evaluation set may contain:

```
complaint_id
domain
language
```

```
intent
text
```

For selected subsets:

```
similar_complaint_id
summary
```

The intent labels will follow the unified ontology.

Two annotators should independently label a substantial subset.

Agreement will be measured using an appropriate inter-annotator agreement metric such as Cohen's kappa for categorical labels.

Disagreements will be reviewed and the annotation guidelines updated before the final evaluation split is frozen.

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# 17. Data Leakage Prevention

Because cross-domain transfer is the central research question, dataset leakage is a major concern.

The project will explicitly prevent:

- duplicate complaints across train/test sets;
- near-duplicate paraphrases across splits;
- translated versions of the same complaint appearing in different splits;
- source-specific leakage;
- label leakage through metadata;
- synthetic examples leaking into the evaluation set.

The final test set will be frozen before final model comparison.

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# 18. Baselines

The project will use progressively stronger baselines.

### Traditional NLP

**TF-IDF + Logistic Regression**

**TF-IDF + SVM**

### Transformer baselines

**BERT**

**mBERT**

### Proposed encoder

**XLM-R**

### Proposed research variant

**XLM-R + contrastive learning**

This allows the paper to distinguish improvements from:

- traditional representations;
- monolingual transformers;
- multilingual pretraining;
- task-specific fine-tuning;
- contrastive representation learning.

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# 19. Evaluation Framework

## Language Identification

Metrics:

- Accuracy
- Macro-F1
- Per-language precision/recall/F1
- Confusion matrix

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## Intent Classification

Primary:

**Macro-F1**

Additional:

- Accuracy
- Precision
- Recall
- Per-class F1
- confusion matrix

The most important comparison will be:

```
In-domain F1
vs.
Cross-domain F1
vs.
Cross-language F1
```

---

## Cross-Domain Transfer

Report:

```
Training domains
Held-out domain
Number of training examples
Number of test examples
Macro-F1
Per-class F1
```

Results will be reported for multiple held-out domains where data permits.

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## Semantic Retrieval

Metrics:

- Precision@5
- Recall@5
- MRR

Baselines:

- TF-IDF cosine similarity
- XLM-R embeddings
- XLM-R + contrastive learning

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## Summarization

Automatic:

- ROUGE-1
- ROUGE-2
- ROUGE-L
- BERTScore

Human:

- factuality
- completeness
- usefulness
- conciseness

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## Explainability

Evaluation will focus on:

- token attribution analysis
- deletion-based faithfulness
- explanation consistency
- qualitative error analysis

No improvement percentage will be claimed without experimental evidence.

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# 20. Ablation Studies

The final paper should contain controlled ablations.

### Ablation 1

XLM-R without contrastive learning

vs.

XLM-R with contrastive learning

### Ablation 2

English-only training

vs.

multilingual training

### Ablation 3

No Hinglish data

vs.

Hinglish included during adaptation

### Ablation 4

Single-domain training

vs.

multi-domain training

### Ablation 5

Standard lexical retrieval

vs.

XLM-R retrieval

### Ablation 6

Different amounts of target-domain data

```
0
25
50
100
250
```

This will allow the paper to answer whether cross-domain transfer remains useful when only a small number of examples from a new domain are available.

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# 21. Error Analysis

A dedicated error-analysis section will examine:

### Language errors

- Hindi incorrectly detected as Hinglish
- Hinglish incorrectly detected as English
- transliteration variation

### Domain errors

- similar intents occurring across different domains
- domain-specific terminology

### Intent errors

- semantically overlapping categories
- ambiguous complaints

- multi-intent complaints

### Cross-domain errors

- domain-specific language not observed during training
- unseen intent vocabulary
- differences in annotation conventions

### Summarization errors

- missing key information
- incorrect interpretation
- unsupported information
- excessive compression

### Explainability errors

- misleading token attribution
- unstable explanations
- irrelevant high-attribution tokens

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# 22. Final Application

The final system will expose a simple API and interface.

### Input

```
Mera recharge successful hai but internet abhi tak
activate nahi hua.
```

### Output

```
Language:
Hinglish

Domain:
Telecom

Intent:
Service Activation Failure

Confidence:
[model output]

Similar Complaints:
1. ...
2. ...
3. ...
```

```
Summary:
[generated summary]

Explanation:
[SHAP-based evidence]
```

The interface will also allow the user to inspect the retrieved similar complaints.

The system is intended as a **research demonstration**, not as a production grievance-management platform.

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# 23. Technology Stack

## Machine Learning

- Python
- PyTorch
- Hugging Face Transformers
- Hugging Face Datasets
- scikit-learn
- Sentence/embedding utilities where required
- SHAP

## Models

- XLM-R
- mT5
- lightweight language-ID model

## Retrieval

- FAISS

## Backend

- FastAPI
- Uvicorn

## Frontend

Initial prototype:

- Streamlit or equivalent lightweight interface

Optional final interface:

- React

### Storage

- PostgreSQL if persistent application storage is required
- FAISS for semantic retrieval

A separate vector database such as Milvus is not required for the expected project-scale dataset.

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# 24. Infrastructure

The available AI Node is suitable for this project.

The supplied infrastructure provides:

- 2 NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs
- 16 MIG instances of 1g.24gb
- approximately 306 GB RAM
- GPU-enabled PyTorch
- CUDA
- NVIDIA NeMo
- vLLM/TGI
- Slurm
- Kubernetes

The project does not require all available GPU resources simultaneously.

The GPU infrastructure will primarily be used for:

- XLM-R fine-tuning
- contrastive-learning experiments
- multilingual summarization experiments
- batch embedding generation
- evaluation
- parallel model experiments

FAISS retrieval can initially run on CPU because the expected project dataset is far below the scale at which GPU indexing becomes necessary.

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# 25. Reproducibility

Every experiment will record:

- dataset version
- preprocessing version
- model checkpoint
- hyperparameters
- random seed
- training configuration

- train/validation/test split
- evaluation script version

The final repository should contain:

```
data/
src/
models/
configs/
experiments/
evaluation/
api/
ui/
notebooks/
docs/
```

Where dataset licenses permit, dataset preparation scripts will be released rather than redistributing restricted datasets.

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# 26. 16-Week Development Plan

## Phase 1 — Data and Foundations

### Weeks 1–3

### Week 1

- audit public datasets
- verify licenses
- identify domains
- inspect language distributions
- define initial intent ontology
- establish repository
- configure Slurm/GPU environment

**Deliverables:**

- dataset inventory
- data-access report
- initial ontology
- reproducible environment

### Week 2

- download/prepare approved public datasets
- deduplicate
- normalize text
- identify candidate training/test sources

- build preprocessing pipeline

**Deliverables:**

- cleaned dataset v1
- preprocessing scripts
- duplicate-detection report

### Week 3

- begin project-specific annotation
- implement language-ID baseline
- establish annotation guidelines
- create train/validation/test split strategy

**Deliverables:**

- annotation guidelines
- LID baseline
- first annotated subset

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# Phase 2 — Core Complaint Understanding

### Weeks 4–7

### Week 4

Implement:

- TF-IDF + Logistic Regression
- TF-IDF + SVM

**Deliverables:**

- baseline results
- baseline evaluation scripts

### Week 5

Implement:

- mBERT baseline
- XLM-R baseline

**Deliverables:**

- transformer training pipeline
- initial model checkpoints

### Week 6

Fine-tune XLM-R for complaint intent.

**Deliverables:**

- trained XLM-R classifier
- in-domain results
- per-domain error analysis

### Week 7

Conduct first cross-domain experiments.

**Deliverables:**

- leave-one-domain-out results
- cross-domain confusion matrices
- preliminary transfer analysis

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# Phase 3 — Research Component

### Weeks 8–10

### Week 8

Implement contrastive-learning setup.

Create:

- positive pairs
- multilingual pairs
- paraphrase pairs
- hard negatives

**Deliverables:**

- contrastive training pipeline
- pair-generation methodology

### Week 9

Train and compare:

```
XLM-R
vs.
XLM-R + Contrastive Learning
```

**Deliverables:**

- comparison results
- embedding analysis

### Week 10

Implement semantic retrieval.

**Deliverables:**

- FAISS index
- top-k retrieval API
- TF-IDF vs XLM-R retrieval comparison

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# Phase 4 — Supporting Components

### Weeks 11–12

### Week 11

Implement multilingual summarization.

**Deliverables:**

- mT5 baseline
- complaint-summary evaluation set
- automatic evaluation

### Week 12

If feasible:

- fine-tune/adapt summarization model
- human evaluation
- implement SHAP

**Deliverables:**

- final summarization pipeline
- SHAP explanation pipeline
- explanation evaluation

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# Phase 5 — System Integration

### Weeks 13–14

### Week 13

Build:

- FastAPI backend
- inference pipeline
- model loading
- FAISS retrieval
- summary generation
- SHAP output

### Week 14

Build:

- lightweight UI
- complaint input
- result display
- similar-complaint visualization
- explanation display

**Deliverable:**

Working end-to-end prototype.

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# Phase 6 — Final Evaluation and Paper

### Weeks 15–16

### Week 15

Run final:

- cross-domain experiments
- cross-language experiments
- ablations
- retrieval experiments
- summarization evaluation
- explainability evaluation
- error analysis

Freeze the final results.

**Week 16**

Produce:

- final technical report
- research paper draft
- architecture documentation
- reproducibility package
- demonstration video
- final presentation
- GitHub repository

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# 27. Final Deliverables

## Research

- [ ] One complete research paper
- [ ] Literature review
- [ ] Defined research questions
- [ ] Defined hypotheses
- [ ] Unified complaint-intent ontology
- [ ] Cross-domain experimental protocol
- [ ] Cross-language experimental protocol
- [ ] Contrastive-learning experiment
- [ ] Ablation studies
- [ ] Error analysis
- [ ] Reproducible evaluation scripts

## Dataset

- [ ] Public-dataset inventory
- [ ] Dataset licensing documentation
- [ ] Unified preprocessing pipeline
- [ ] Deduplication pipeline
- [ ] Project-specific Indian complaint dataset
- [ ] Annotation guidelines
- [ ] Inter-annotator agreement analysis
- [ ] Frozen test set
- [ ] Dataset documentation

## Models

- [ ] Language-ID model
- [ ] TF-IDF baselines
- [ ] mBERT baseline
- [ ] XLM-R baseline
- [ ] XLM-R contrastive model
- [ ] Semantic retrieval index

- [ ] mT5 summarization pipeline
- [ ] SHAP explanation module

## Application

- [ ] FastAPI backend
- [ ] Complaint-analysis endpoint
- [ ] Semantic retrieval endpoint
- [ ] Summarization endpoint
- [ ] SHAP explanation endpoint
- [ ] Lightweight web interface
- [ ] End-to-end demonstration

## Engineering

- [ ] Reproducible environment
- [ ] Configuration files
- [ ] Training scripts
- [ ] Evaluation scripts
- [ ] Experiment logs
- [ ] Model checkpoints
- [ ] API documentation
- [ ] Installation documentation
- [ ] Usage examples

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# 28. Success Criteria

The project will **not promise specific model scores before experimentation**.

Instead, success will be evaluated according to whether the project can answer the research questions convincingly.

### Minimum technical success

The project should produce:

1. A functioning multilingual complaint classifier.
2. A measurable cross-domain evaluation.
3. A measurable cross-language evaluation.
4. A comparison between standard XLM-R and contrastive XLM-R.
5. A functioning semantic retrieval component.
6. A functioning multilingual summarizer.
7. A functioning SHAP explanation component.
8. A reproducible end-to-end system.

### Research success

The project should establish experimentally whether:

- multilingual representations transfer across domains;
- contrastive learning improves transfer;
- multilingual training improves Hindi/Hinglish performance;
- learned embeddings improve semantic retrieval;
- summarization is sufficiently useful for complaint triage;
- SHAP explanations provide faithful evidence for predictions.

A negative result is still scientifically useful if the experiments are properly controlled and analyzed.

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# 29. Scope Boundaries

The following are explicitly **out of scope**:

- LLM-based reasoning
- RAG
- NER
- urgency/severity prediction
- systemic-failure detection
- temporal anomaly detection
- automated complaint resolution
- autonomous responses to complainants
- production integration with government/banking systems
- medical/legal decision-making
- training a language model from scratch

These may be listed as future extensions, but they will not be required for project completion.

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# 30. Expected Research Contribution

The project does not assume that it will establish a new state of the art.

Its intended contribution is an empirical study of:

1. **Cross-domain complaint intent transfer**
2. **Multilingual and code-mixed complaint understanding**
3. **Contrastive multilingual complaint representations**
4. **Semantic retrieval of related complaints**
5. **A small, carefully documented Indian multilingual complaint evaluation resource**
6. **A reproducible end-to-end complaint-understanding system**

The strength of the paper will depend on the quality of the experimental design, dataset construction, baselines, ablations, and analysis rather than on claiming a predetermined performance improvement.

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# 31. Publication Strategy

## Primary Target — ACL 2027

**Association for Computational Linguistics Annual Meeting**

ACL 2027 will be held in Kyoto, Japan, August 17–22, 2027. As of the current project planning date, the official site lists the ARR submission, commitment, and acceptance dates as **TBA**.

This is the preferred target because the project is fundamentally an NLP representation-learning and transfer-learning study.

**Target paper type:** Long paper if the final experiments support a sufficiently substantial contribution; otherwise short paper.

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## Secondary Target — COLING 2027

COLING 2027 will be held in Macau, May 9–14, 2027.

Its official call explicitly includes benchmarking/evaluation, multilingual NLP-related areas, semantics, and NLP applications, making it a reasonable alternative venue for this work.

However, its ARR submission deadline is **October 12, 2026**. Therefore, submitting this project to COLING 2027 would require an accelerated paper-development schedule and should **not** be assumed as the default route.

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## Secondary Target — NAACL 2027

NAACL 2027 will take place June 1–5, 2027.

The official call lists an ARR submission deadline of **October 12, 2026**, with commitment on December 23, 2026.

The project fits the conference's stated scope of empirical, methodological, resource-oriented, evaluation, and reproducibility contributions.

Again, the October deadline makes it an **accelerated option**, not the default assumption.

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## Not a Current Target — EACL 2027

EACL 2027 is not included in the primary submission plan because its ARR submission deadline was August 3, 2026, which has already passed relative to this project's August 2026 start.

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# 32. Publication Positioning

The paper should **not** be presented as:

> "A complete AI complaint-management platform."

It should be presented as:

> **An empirical study of multilingual complaint representations and their ability to transfer across domains, languages, and limited-data settings.**

The deployed system is the demonstration vehicle.

The paper's core contribution is the **experimental investigation of representation transfer**.

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# 33. Final Project Definition

## Title

**Cross-Domain Complaint Understanding via Multilingual Encoders**

## Core model

**XLM-R**

## Research extension

**Contrastive learning**

## Language component

**English / Hindi / Hinglish language identification**

## Primary NLP task

**Complaint-intent classification**

## Secondary NLP task

**Semantic retrieval of related complaints**

## Generation component

**mT5-based complaint summarization**

## Explainability

**SHAP-based intent explanation**

## Retrieval

**FAISS**

## Backend

**FastAPI**

## Interface

**Lightweight web application**

## Research focus

**Cross-domain + cross-language transfer under limited labeled data**

## Primary publication target

**ACL 2027**

## Secondary publication targets

**COLING 2027 / NAACL 2027**, subject to submission timing and experimental maturity.

## Explicitly removed

**LLM, RAG, NER, urgency prediction, anomaly detection, systemic-failure clustering, and autonomous resolution.**

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## Final Architecture

```
                ┌───────────────────┐
                │     COMPLAINT     │
                └─────────┬─────────┘
                          │
                          ▼
             ┌───────────────────────┐
             │ Language Identifier   │
             │ EN / HI / Hinglish    │
             │ / Other               │
             └──────────┬────────────┘
```

```
                    │
                    ▼
          ┌───────────────────────┐
          │      XLM-R Encoder    │
          │                       │
          │ Multilingual +        │
          │ Cross-domain learning │
          └───────────────────────┘
                    │
                    │
       ┌────────────┼─────────────┐
       │            │             │
       ▼            ▼             ▼
┌────────────┐ ┌────────────┐ ┌────────────┐
│   Intent   │ │  Semantic  │ │ Summarizer │
│ Classifier │ │  Retrieval │ │    mT5     │
└────────────┘ └────────────┘ └────────────┘
      │             │              │
      ▼             ▼              ▼
    Intent     Similar Cases    Summary
      │
      ▼
┌────────────────┐
│      SHAP      │
│   Explanation  │
└────────────────┘
       │
       ▼
┌────────────────────────────┐
│       Final Output         │
│                            │
│ Language                   │
│ Domain / Intent            │
│ Confidence                 │
│ Similar Complaints         │
│ Complaint Summary          │
│ SHAP Explanation           │
└────────────────────────────┘
```

**This is the version I would freeze for faculty approval.** It is ambitious enough to be a serious NLP research project, but the scope is now controlled enough that the team can actually finish the research, the working system, and one defensible paper rather than producing seven partially completed modules.